

The reflected ridge

The (β_0, λ, z) non-identifiability behind the residual time-block explosions

Technical note

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Abstract

Removing `z.init = 0.01` reduced the exploding time-block forecasts from 21/25 blocks to 4/25 (setting 4, AR(2), datasets 1–5). This note diagnoses the residual 4 from the 25 retained refits of the diagnostic harness (`fit.diagnostics.R`). The finding is a single empirical law: across *all* 25 blocks — healthy and exploded alike — the maximum predictive standard deviation is proportional to the fitted skew loading, $\max_h \text{sd}(\hat{y}_h) \approx 1.49 |\hat{\lambda}|$, with $R^2 = 0.987$ through the origin. There is no separate “explosion” mechanism: the flagged blocks are simply the ones whose chain settled at an inflated $|\hat{\lambda}|$. The inflation lives on an exact likelihood ridge — the data see (β_0, λ, z_t) only through $m_t = \beta_0 + \lambda z_t$ — and the failing chains occupy a *reflected* branch of it: λ flips sign to -10 to -15 , β_0 rises to 26–35 so that $\beta_0 + \lambda \bar{z}$ still equals the data mean, the marginal scale of z stays correct (the fit *passes* the $\bar{z}$ self-consistency assertion), and ϕ_z drifts to a near-unit root because the reflected z path must be smooth. The forecast then redraws z from its half-normal law at full temporal variance and multiplies by the inflated $|\lambda|$. This is the same ridge that the `z.init` bug exploited, entered through a different door, and it is a property of the model, not of any one line of code.

1 Where things stand

After the `z.init` fix, the 25 blocks of setting 4 (truth: $\lambda = 3$, $\beta_0 = 10$, $\phi_z = (0.80, -0.35)$, data marginal SD 4.38) were refitted with the diagnostic harness, which retains per-fit parameter summaries and runs three assertions:

- A** z self-consistency: $\bar{z} \approx \mathbb{E}[\tau^{-1/2}] \sqrt{2/\pi}$ (the model *is* $z \sim \text{HN}(0, \tau^{-1/2})$);
- B** truth recovery: $\hat{\lambda}$ has the sign and rough size of $\lambda = 3$, $\hat{\beta}_0 \approx 10$;
- C** predictive spread: $\max_h \text{sd}(\hat{y}_h) \leq 3 \times$ the data marginal SD (a stationary AR forecast converges to climatology).

Four blocks fail C. Seven blocks — the four C-failures plus three milder cases — share one signature, shown in Table 1.

Three regularities, all of which the rest of this note explains:

1. every anomalous block has $\hat{\lambda} < 0$ with $\hat{\beta}_0$ raised by almost exactly $-\hat{\lambda}\bar{z}$, so the fitted mean is untouched;
2. $\bar{z}$ is *healthy* in every one of them — assertion A passes; this is **not** the frozen- z failure of the `z.init` bug;

Table 1: The seven reflected blocks, with one healthy block for contrast. $\mu = \hat{\beta}_0 + \hat{\lambda}\bar{z}$; the data mean is reproduced by every row, healthy or not. Assertion A passes in every row shown.

(dset, blk)	flag	$\hat{\lambda}$	$\hat{\beta}_0$	$\bar{z}$	$\hat{\phi}_{1z}$	$\max_h \text{sd}$	μ	data mean
truth / target		+3	10	≈ 1.5	0.80	≤ 13.1		
(1, 4)	FAIL C	−12.10	34.99	1.66	1.15	20.7	14.9	15.0
(3, 4)	FAIL C	−15.48	29.75	0.98	0.89	20.5	14.5	14.6
(5, 2)	FAIL C	−10.77	26.78	1.09	0.93	14.5	15.1	15.4
(2, 4)	FAIL C	−9.67	26.31	1.22	1.03	14.2	14.5	14.7
(5, 1)	—	−8.00	27.04	1.52	1.00	11.4	14.9	15.2
(5, 4)	—	−6.41	25.99	1.77	0.92	9.4	14.7	14.8
(4, 1)	—	−4.10	18.61	1.48	0.70	6.4	12.6	12.8
(1, 3) healthy		+2.80	10.36	1.68	0.62	5.3	15.1	15.0

3. the four “explosions” are not a species apart: they are the largest $|\hat{\lambda}|$ values on a single continuous law (Figure 1).

2 The ridge, exactly

With one knot ($K = 1$) the observation equation at site s , time t is

$$y_{st} = x'_{st}\beta + \lambda z_t + \varepsilon_{st}/\sqrt{\tau_t}, \quad z_t \sim \text{HN}(0, \tau_t^{-1/2}), \quad (1)$$

so the likelihood involves $(\beta_0, \lambda, \{z_t\})$ *only through*

$$m_t = \beta_0 + \lambda z_t. \quad (2)$$

Fix any fitted trajectory $\{m_t\}$. Then for *every* pair (β_0^*, λ^*) with $\lambda^* \neq 0$, the reassignment

$$z_t^* = \frac{m_t - \beta_0^*}{\lambda^*} \quad (3)$$

reproduces the data likelihood exactly, provided $z_t^* \geq 0$ for all t . The likelihood is *flat* on this two-parameter family. Identification comes entirely from the prior structure on z : its half-normal marginal tied to τ_t , and the AR(2) Gaussian-copula dynamics on $z_t^* = \text{hn.cop}(z_t, \tau_t^{-1/2})$.

Two branches of the family matter here.

The positive branch ($\lambda^* > 0$). z_t^* is an increasing affine map of m_t ; the z path keeps the shape of the common temporal signal. The prior is happiest near the truth, and 18 of 25 chains sit there.

The reflected branch ($\lambda^* < 0$). z_t^* is a *decreasing* affine map of m_t : the latent path is flipped upside down, and β_0^* must exceed $\max_t m_t$ to keep $z_t^* \geq 0$ — which is exactly the observed $\hat{\beta}_0 \approx 26\text{--}35$ against a data mean of ≈ 14.5 . Seven of 25 chains ended here.

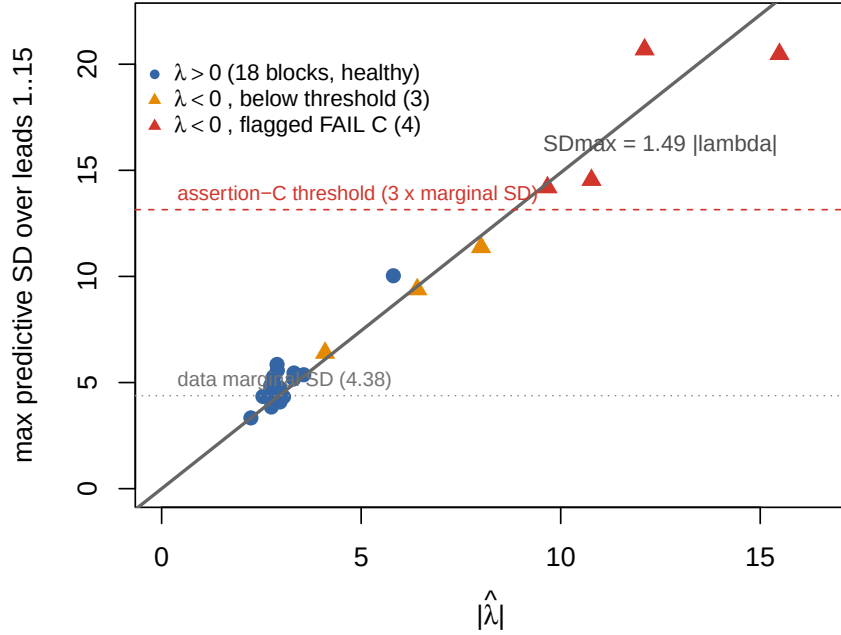


Figure 1: All 25 refitted blocks. The maximum predictive SD over leads 1–15 is proportional to $|\hat{\lambda}|$: slope 1.49, $R^2 = 0.987$ through the origin. Healthy fits ($\hat{\lambda} \approx 3$) and flagged explosions ($|\hat{\lambda}| \approx 10$ –15) lie on the same line; the assertion-C threshold merely cuts the line at $|\hat{\lambda}| \approx 9$. “Fixing the forecast” is therefore not an available move: the forecast is a faithful amplifier of $|\hat{\lambda}|$.

3 Why the reflected branch is nearly free

Three prices must be paid to sit on the reflected branch, and each one turns out to be small.

3.1 The prior on λ charges almost nothing

The prior is $\lambda \sim N(0, \text{lambda.s} = 20)$ (`mcmc_ar2.R`). Whether 20 is a standard deviation or a variance, the log-prior difference between $\hat{\lambda} = -12$ and $\lambda = +3$ is a fraction of a nat to a few nats — negligible against a likelihood built from $\text{ns} \times T$ observations that the branch reproduces *exactly*. A diffuse sign-symmetric prior cannot veto a sign flip that the likelihood does not oppose.

3.2 The half-normal marginal is satisfiable — at the price of smoothness

On the reflected branch, by (3),

$$\bar{z}^* = \frac{\bar{m} - \beta_0^*}{\lambda^*}, \quad \text{sd}_t(z^*) = \frac{\text{sd}_t(m)}{|\lambda^*|}. \quad (4)$$

The *mean* constraint is easy: for any λ^* , choose $\beta_0^* = \bar{m} - \lambda^* \bar{z}^{\text{target}}$ and the fitted $\bar{z}$ lands wherever the half-normal marginal wants it (≈ 1.5 here). This is precisely why **assertion A passes on every reflected fit**: the marginal *location* of z is preserved by construction.

The *temporal variation* is not. Equation (4) says the z path’s variation shrinks like $1/|\lambda^*|$: at $\hat{\lambda} = -12$ the latent path has one-quarter of the temporal variation of the truth-branch path. The model marginal $\text{HN}(0, \tau^{-1/2})$ prescribes $\text{sd}(z) = \sigma\sqrt{1 - 2/\pi} \approx 1.1$; the reflected fit carries a z path that is far too smooth for its own marginal. Note the asymmetry with the `z.init` bug: there, the *level* of z was wrong ($7\times$ too small) and A failed loudly; here the level is right and only the *second moment in time* is wrong — a quantity assertion A never looks at.

3.3 The AR(2) copula absorbs the smoothness

A too-smooth z^* path is not rejected by the AR(2) prior — it is *fitted* by it, with a near-unit root. That is the observed $\hat{\phi}_{1z} = 0.89\text{--}1.15$ (truth 0.80; the joint stationarity constraint keeps $\phi_1 + \phi_2 < 1$, so these are legal, highly persistent configurations). The persistence in turn makes the smooth path likely, closing the loop. This is the same self-reinforcing smoothness that appeared in the `z.init` cascade, but running in the opposite causal direction: there, frozen z dragged ϕ_z to a unit root; here, inflated $|\lambda|$ *requires* a smooth z , and the near-unit-root ϕ_z is what makes smoothness cheap. ϕ_z is the accomplice, not the trigger.

Once inside, the chain is trapped: leaving the branch requires moving λ across zero *and* dropping β_0 by ≈ 20 *and* re-inverting the entire z path simultaneously. Every one-coordinate Gibbs or MH move away from the branch is individually catastrophic for the likelihood, so the sampler stays. The posterior is bimodal; the chain picks a mode early and cannot switch.

4 Why the forecast explodes, quantitatively

`forecast.block` draws z_{fc} from the model’s own law: the AR(2) recursion on the copula scale relaxes to $N(0, 1)$, so z_{fc} relaxes to its $\text{HN}(0, \tau_{\text{fc}}^{-1/2})$ marginal with *full* temporal variance — the forecast does not know the fitted path was abnormally smooth. The skew channel then contributes spread

$$\text{sd}(\lambda z_{\text{fc}}) = |\lambda| \cdot \mathbb{E}[\sigma] \sqrt{1 - 2/\pi} \cdot (1 + \tau\text{-dispersion}) \approx 1.13 |\lambda| \text{ at the floor,} \quad (5)$$

using $\mathbb{E}[\sigma] \approx 1.9$ from the fitted (a, b) . With the τ forecast’s own dispersion and the residual noise folded in, the observed constant is 1.49 (Figure 1). At $\hat{\lambda} = 3$ this gives ≈ 4.5 — climatology, as it must. At $\hat{\lambda} = -15.5$ it gives ≈ 20 against a data marginal of 4.38 — the residual explosions, with nothing else needed to explain them.

The forecaster is again the detector, not the culprit: it is the only stage that draws z from the model law instead of reading the chain’s smoothed path, so it is the only stage where the branch’s hidden inconsistency — inflated $|\lambda|$ against shrunken z variation — becomes visible.

5 Two doors, one ridge

	<code>z.init</code> bug (fixed)	reflected branch (residual)
entry	deterministic: every chain started at a 2.4σ tail point	stochastic: 7/25 chains drift in during burn-in
$\bar{z}$ (assertion A)	$7\times$ too small — fails	correct — passes
what is wrong with z	its level	its temporal variation
$\hat{\lambda}$	-56 to -98	-4 to -15
$\hat{\beta}_0$	25–36	19–35
$\hat{\phi}_{1z}$	$\rightarrow$ unit root (effect)	$\rightarrow$ unit root (enabler)
predictive SD	40–104	6–21, $= 1.49 \hat{\lambda} $
fix	delete one argument	none available at the code level: this is the model’s own geometry

The common object is the ridge (2). The `z.init` bug forced every chain onto it through the z coordinate; with that door closed, a minority of chains still find it through the (λ, β_0) coordinates. The block-4 concentration (3 of the 4 C-failures sit at seam 140) has no identified data-side cause — dataset 4’s own block 4 is perfectly healthy ($\hat{\lambda} = +2.97$) — and is consistent with chance placement of 7 stochastic events on a 5×5 grid, possibly tilted by whatever the seam-140 prefix does to the early posterior on λ . We flag this explicitly as unresolved rather than invent a mechanism for it.

6 What can actually be done

In rough order of cost:

1. **Guard and reseed (pipeline-level, cheap).** The reflected branch is now detectable at fit time from three one-line symptoms: $\hat{\lambda} < 0$ (truth known in simulation), $\hat{\beta}_0$ far above the data mean, $\text{sd}_t(\hat{z})$ far below $\hat{\sigma}\sqrt{1 - 2/\pi}$. Refit flagged blocks with a different seed until clean. This is honest — the assertion is a model-consistency check, not an outcome filter — but it treats the symptom and would bias nothing only as long as the clean mode is the dominant one.
2. **A new assertion A' (diagnostic-level, cheap).** Assertion A checks the first moment of z ; the reflected branch is invisible to it. Add A': $\text{sd}_t(\hat{z}_t) \approx \hat{\sigma}\sqrt{1 - 2/\pi}$, the *second* temporal moment the model prescribes. Prediction, testable on the next diagnostic run: reflected fits violate A' by roughly the factor $|\hat{\lambda}|/3$, healthy fits do not.
3. **Break the sign symmetry in the prior (model-level, moderate).** In simulation the sign of λ is known; even in applications the sign of skewness is usually known from the data (the backend itself initialises λ at the sample skewness). A prior with mass on one side (e.g. $\lambda \sim \text{HN}$, or $N(3, \text{small})$ in simulation) removes the reflected branch outright. This is the standard resolution of the same sign-reflection non-identifiability in skew-normal / factor-loading models.
4. **Tighten `lambda.s` (model-level, blunt).** $N(0, 20)$ charges a few nats at most for $|\lambda| = 12$. A scale chosen so that $|\lambda| \approx 12$ is genuinely expensive would starve the reflected

branch without forbidding negative λ — but it also shrinks honest estimates, and picking the scale is a judgement call.

Option 3 is the principled fix; options 1–2 are the immediate ones, require no backend change, and are sufficient to produce the first trustworthy full time-block study.

7 The lesson this bug adds

The `z.init` note ended by prescribing self-consistency assertions; this episode is their first live test, and the verdict is double-edged. Assertion A worked exactly as designed — by *passing*, it killed the frozen- z hypothesis on contact and forced the diagnosis toward the ridge. Assertion C worked as designed — it found every case worth finding, and Figure 1 shows it is a threshold on a clean underlying law rather than an arbitrary flag. But the reflected branch slipped past A because A checks the one moment of z that the branch preserves *by construction*.

An identifiability failure will satisfy every check that lies along the ridge. Assertions must be chosen *transverse* to the ridge: here, the level of z is along it, and the temporal variation of z is across it.

Concretely: a fit can reproduce the data mean, pass its own marginal-location check, and carry legal AR coefficients, while being a reflection of the truth with a four-fold inflated loading. The only quantities that betray it are the ones the likelihood does not pin — and those are exactly the quantities a forecast consumes.